INCH-POUND

A-A-54796
11 January 1993

COMMERCIAL ITEM DESCRIPTION

HYDROMETER, GRADUATED SCALE
(Specific Gravity Range 1.196 to 1.286)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-H-37082 which is cancelled.

This Commercial Item Description covers a Type 80/60 graduated scale hydrometer for determining end-of-charge specific gravities of submarine lead acid storage batteries.

Salient characteristics:

Shall be a hydrometer conforming to Figure 1 for style and design. The hydrometer shall be suitable for determining end-of-charge specific gravities of submarine lead acid storage batteries in the range of 1.196 to 1.286, and in 0.002 increments.

Shall be a type 80/60 hydrometer that is graduated to read specific gravity at 80 degrees F referred to water at 60 degrees F.

Shall consist of the following:

A glass stem.

A glass body.

A graduated scale securely affixed inside the stem.

Lead shot ballast that fills the lower end of the glass body.

AMSC N/A FSC 6630

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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The lead shot ballast shall be held securely in place within the lower end of the glass body with a sealant that does not soften below 200 degrees F. As indicated in Figure 1, the glass body shall include four evenly spaced nubs that are located approximately 3/4 inches below the top of the body, and four evenly spaced nubs that are located approximately 3/4 inches above the bottom of the body. The glass used for the stem and body shall conform to the applicable requirements in ASTM E100.

Shall include a graduated scale conforming to Figure 1. The material for the graduated scale shall conform to the applicable requirements in ASTM E100. The 5 point dotted lines for the scale in Figure 1 shall be marked in red and the other lines shall be marked in black. As an alternate, the red and black lines denoted in Figure 1 as being dotted or dash lines may be supplied as solid lines. The scale shall be inserted in the stem so that all numerical markings and the entire range of 1.196 to 1.286 are visible. The higher end of the scale range shall be inserted towards the glass body. The scale within the stem shall be straight and shall be anchored by a design which prevents it from moving. As an alternate, the scale may be fixed in place by a sealant that does not soften below 200 degrees F and does not deteriorate with time.

Shall provide the graduated scale with an accuracy of $\pm$ 0.0015 specific gravity units based on verification testing in accordance with ASTM E126 (see Quality Assurance Provisions).

Shall provide the hydrometer with the following dimensions:

An overall length of 7 to 7-9/16 inches.

An overall body diameter (including nubs) of 0.650 to 0.720 inches.

An outside stem diameter of 0.155 to 0.170 inches.

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Shall provide hydrometer with identification marking that includes the following:

Specific gravity temperature ($80^{o}/60^{o}F$).

Units of hydrometer scale.

Manufacturer's name or registered trademark.

Manufacturer's part number.

The marking shall be legible and permanent and shall be positioned in a readibly visible location on the inside of the hydrometer.

Workmanship. The hydrometer shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One hydrometer, as specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Inspections and tests. Inspections and tests shall be conducted in accordance with commercial practice to determine compliance with specification requirements. Sampling for the inspections and tests shall be conducted in accordance with commercial practice. In addition, the accuracy of the graduation markings shall be determined in accordance with the requirements herein after indicated (see "Hydrometer graduated scale testing").

Hydrometer graduated scale testing. The accuracy of the hydrometer scale shall be determined in accordance with ASTM E126. The testing shall include four points on the hydrometer scale that are at least 0.02 specific gravity units from each other. Sampling for the accuracy testing shall be conducted in accordance with MIL-STD-105 for inspection level S-1. A hydrometer with any reading that is not within the required ± 0.0015 specific gravity units for acceptable accuracy shall be cause for rejection. The acceptance number for the sampling plan shall be zero.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

OTHER PUBLICATIONS.

ASTM

ASTM E100	Standard Specification for ASTM Hydrometers.
ASTM E126	Standard Method for Inspection and Verification of Hydrometers.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

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NOTE: The following National Stock Number is covered by this document:

NSN	Item Identification
6630-00-580-3951	HYDROMETER, GRADUATED SCALE, Specific Gravity Range 1.196 to 1.286

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

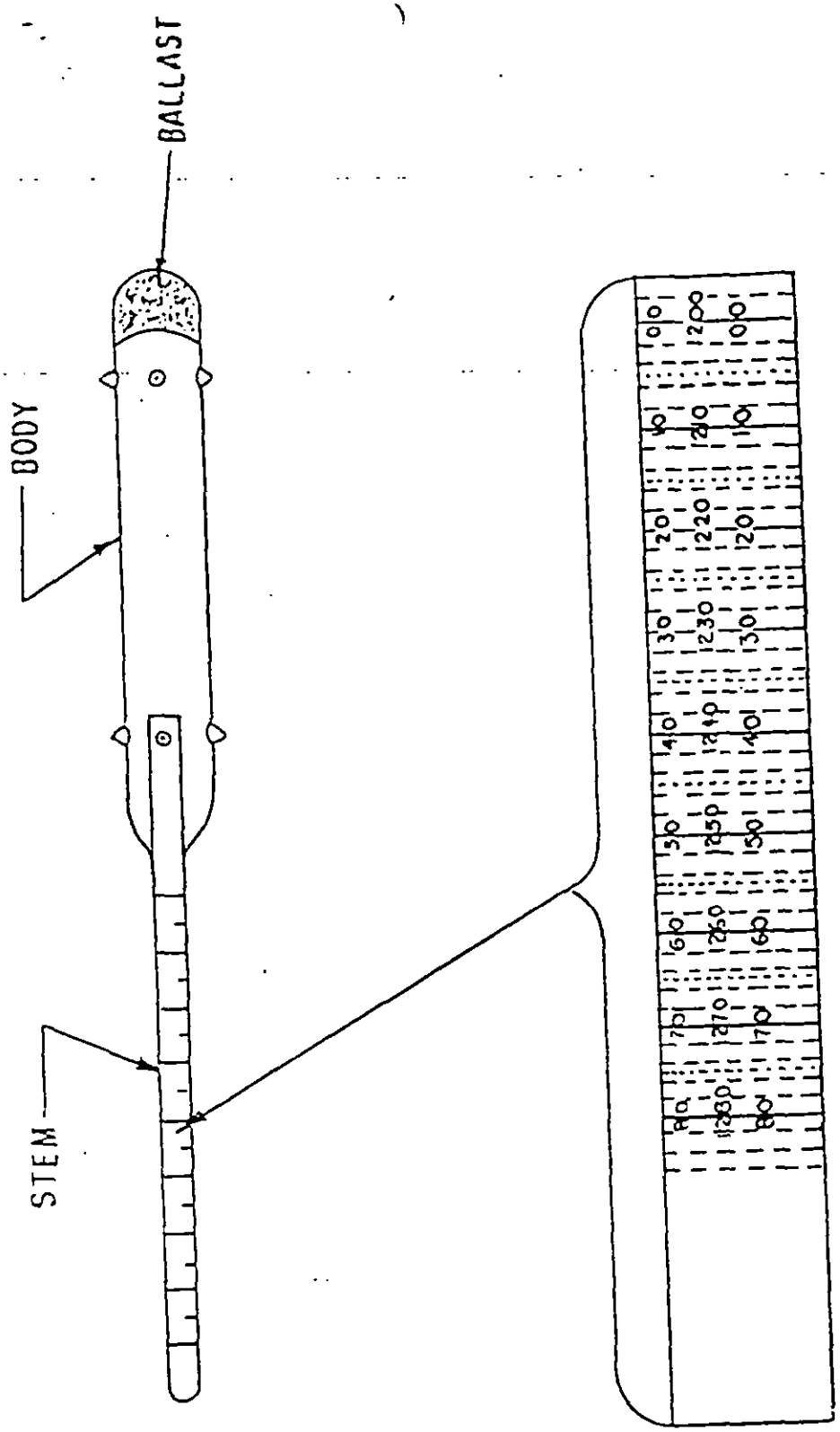
DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6630-0533

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RANGE: 1.196 TO 1.286

GRADUATED SCALE HYDROMETER

FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54796	2. DOCUMENT DATE (YYMMDD) 11 JANUARY 1993

3. DOCUMENT TITLE
HYDROMETER, GRADUATED SCALE (Specific Gravity Range 1.196 to 1.286)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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